

10 (**short**) A design for a strut requires that it shall carry an axial load without buckling (the Euler buckling load is $F = \pi^2 EI / L^2$ where E is Young's modulus, I is the second moment of area and L is the length). The strut has a fixed length and a hollow, thick-walled circular section.

(a) The thick-walled circular tube has an outer radius of $5t$ and a wall thickness of $2t$. Show that the shape efficiency factor ϕ_B^e for such a strut is approximately equal to 2 using

$$\phi_B^e = \frac{I}{I_{ref}} = \frac{12I}{A^2}$$

where A is the cross-sectional area and I_{ref} is a reference second moment of area of a solid square of the same cross-sectional area. [5]

(b) Taking the cross-sectional area to be a free variable, show that, in order to minimise mass, the following shape-dependent performance index M should be maximised

$$M = \frac{\sqrt{E\phi_B^e}}{\rho}$$

where ρ is the density. Using the data in Table 1, choose (without detailed calculation) between a steel circular tube and a wooden square, based on this performance index.

Material	Young's modulus (GPa)	Density (Mg m ⁻³)
Steel	210	7.8
Wood	20	0.8

Table 1

[5]